

Motherboards and Architecture -- Guided Notes ANSWER KEY

Unit 2.2 | CompTIA Network+ N10-009 Obj. Core 1 3.5 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

Match each motherboard form factor to its dimensions:

ATX: _____ mm x _____ mm

Micro-ATX: _____ mm x _____ mm

Mini-ITX: _____ mm x _____ mm

A smaller motherboard will/will not fit in a larger ATX case.

Answer: ATX: 305 x 244 mm. Micro-ATX: 244 x 244 mm. Mini-ITX: 170 x 170 mm. A smaller motherboard WILL fit in a larger ATX case (using fewer standoffs).

Question 2

Socket LGA 1700 is compatible with Intel _____ through _____ generation Core processors.

Socket AM4 is compatible with AMD Ryzen _____ through _____ series. A CPU from one socket family _____ (will/will not) fit a motherboard with a different socket.

Answer: 12th through 14th generation (Alder Lake, Raptor Lake). AM4 = Ryzen 1000 through 5000 series. Will NOT -- socket incompatibility is physical.

Question 3

DDR4 and DDR5 RAM modules are physically _____ with each other. A DDR5 stick _____ (will/will not) physically fit in a DDR4 slot because the _____ (alignment notch) is in a different position. You must match the RAM generation to the _____ specification.

Answer: Incompatible; will NOT; key notch; motherboard's.

Question 4

By default, RAM operates at its base _____ (e.g., 4800 MHz for DDR5-4800), not its advertised rated speed. To enable the rated speed, you must enable _____ (Intel) or _____ (AMD) in the UEFI settings. Without this, faster RAM sticks run slower than their marketed specification.

Answer: JEDEC speed; XMP (Extreme Memory Profile); EXPO (Extended Profiles for Overclocking).

Question 5

Modern motherboards use _____ firmware instead of the legacy BIOS. UEFI supports the _____ partition table, which allows drives larger than _____ TB and up to 128 primary partitions. Legacy BIOS uses _____ which limits drives to 2 TB.

Answer: UEFI; GPT (GUID Partition Table); 2 TB; MBR (Master Boot Record).

Question 6

Secure Boot is a UEFI feature that checks the _____ signature of the OS bootloader before allowing it to run. Its purpose is to prevent _____ (malware that infects the boot process) from loading. Secure Boot must be enabled for Windows _____ installation.

Answer: Digital (cryptographic) signature; bootkits / rootkits; 11.

Question 7

The CMOS battery (typically a _____ coin cell) powers a small chip that stores the BIOS/UEFI settings when the system is unplugged. If this battery dies, the system will _____ its BIOS settings and reset the _____. Clearing the CMOS (by removing the battery or using the CLRRTC jumper) resets all BIOS settings to _____.

Answer: CR2032; lose; date and time; factory defaults.

Question 8 -- Real World Application

REAL WORLD: A customer bought a motherboard (LGA 1700, DDR5 slots) and plans to install a 13th Gen Intel Core i7, two sticks of DDR5-6000 RAM, and a 4 TB NVMe SSD. They want to run Windows 11. List every UEFI setting they should verify before and after installing Windows 11, and explain what could go wrong if each setting is wrong.

Answer: 1) XMP/EXPO -- must be enabled for DDR5-6000 (without it, RAM runs at DDR5-4800 JEDEC default, wasting performance). 2) Secure Boot -- must be enabled for Windows 11 (disabled = install blocked or system flags unsupported). 3) Boot order -- set NVMe as primary boot device (wrong order = boots wrong drive or fails to boot). 4) UEFI mode -- must not be in Legacy/CSM mode (CSM mode uses MBR, which cannot address drives larger than 2 TB). 5) TPM 2.0 -- must be enabled (required for Windows 11; without it installation wizard blocks the install).